ZOO-502
Animal Physiology and Behavior
Solved Repeated Questions 2018

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Mcqs:
The human ear can detect sound frequencies lying between 20 to 20000 hertz.
In small diameter, nonmyelinated fibers, speedmay be as low as 0.25m/s.
Acclimatization Happens due to chronic exposure to new, naturally occurring environmental conditions.
hyperthyroidism leads to:
high body temp, profuse sweating, weight loss, Irritability high blood pressure.
Zoo 502

Differentiate between Norepinephrine and Epinephrine?2
Norepinephrine is the primary excitatory transmitter in postganglionic cells of sympathetic system.
Epinephrine is excitatory at some synapses and inhibitory at others depending on postsynaptic membrane
2 basic types of stimuli 2 marks
• ***External stimuli***
Stimuli of odor, touch, light, sound and gravitation
• ***Internal stimuli***
Stimuli of pain, homeostatic imbalances and blood pressure
Define vestibular apparatus?
Vestibular apparatus, located in the inner ear. It has:
Saccule
Utricle
Semicircular canals
Name two components of rhdopsin? 2mks
A rhodopsin molecule consists of two major components:

Opsin protein
A light-absorbing molecule (retinal or 3dehydroretinal)

Function of LH in male and female.

In Females LH stimulates production of estrogen and progesterone by ovary.
It causes ovulation and formation of corpus luteum.
In males, LH is also called as ICSH (interstitial cell stimulating hormone) It stimulates testosterone production by the Leydig cells of testes.

What are gonadostropin

Gonadotropins: Luteinizing and Follicle Stimulating Hormones. Luteinizinghormone (LH) and follicle-stimulating hormone (FSH) are called gonadotropinsbecause stimulate the gonads - in males, the testes, and in females, the ovaries.

Define the terms gland and secretion? 2mks

A ***gland*** is a cell or group of cells that secretes a particular chemical substance for use in the body or for discharge into the surroundings.
Secretions are chemical substances, synthesized by glandular cells. They are released from the gland in response to an appropriate stimulus.

Write the role of Fibers of Zonula and ciliary muscles in eye? 3

Fibers of zonula held the lens in place.
They change the shape of lens by exerting an outward tension on perimeter.
Attached with fibers of zonula are the ***ciliary muscles***
They contract to pull fibers of zonula, which flatten the lens. This focuses distant objects on retina. When they relax, lens becomes rounded and near object is focused.

Describe Hodgkin cycle in detail? 3

A stimulus causes local depolarization of the membrane.
Na+ channels open in response to initial depolarization.
This increases Na+ conductance that allows Na+ to enter the cell, which cause further depolarization of membrane and decrease in Vm.
This relationship between Na+ conductance (gNa) and Vm represents a type of positive feedback system and is termed as the Hodgkin cycle.
It results in triggering of an action potential.

Which hormones are involved in the regulation of water and ions? 3

These hormones include:

- Antidiuretic hormone (ADH)
- Aldosterone
- Atrial natriuretic Hormone
- Calcitonin
- Parathormone

Explain two types of feedback mechanism in detail? 5

Feedback controls regulate particular variables e.g.,
Temperature • Salinity • pH

This regulation requires:
Continuous sampling of controlled variables
Respective corrective actions.
Types of Feedback Systems
Two Types:
1. Negative feedback systems
2. Positive feedback systems

Negative Feedback Systems

The most common form of regulation, The end product works to stop or slow down the process
Example-1
Breakdown of sugar in the cells generates ATP, Excess ATP "feeds back" and inhibits an enzyme near the beginning of pathway. ATP production stops.
Example-2
Control of blood sugar (glucose) by insulin; Receptors sense a rise in blood sugar. Control center (pancreas) secretes insulin and Insulin lowers blood sugar levels. As blood sugar level reaches homeostasis, pancreas stops releasing insulin.

Positive Feedback Systems

An end product speeds up the process of its production.
Examples
1. Clotting of blood in response to an injury
2. Release of Oxytocin during labor

Enlist five functions of estrogen hormones in pregnancy? 5

- During pregnancy, estrogens exert mainly a proliferative function on reproductive and associated organs of the mother.

They cause:

- enlargement of uterus
- Enlargement of breasts and growth of breast ductal structure
- Enlargement of external genitalia
- Relaxation of pelvic ligaments to make pubic symphysis elastic for easy passage of the fetus through birth canal

Abnormalities due to GH?5

Disturbances in the secretion of GH lead to abnormal growth and development in humans:

Gigantism

Hypersecretion of GH before puberty causes excessive size and stature.

Acromegaly

Hypersecretion of GH after maturity causes enlargement of bones of the head and limbs.

Dwarfism

Insufficient secretion of GH during childhood and adolescence causes underdevelopment of the body (short stature).

Messiner corpuscales and paccin corpuscales difference?3

Touch receptors in ridges of fingertips are ***Meissner's Corpuscles***. They have encapsulated nerve endings.

To receive pressure stimulus, ***Pacinian corpuscles*** are situated deep in skin, in the limbs they receive vibrations.

Oxytocin? 3

In mammals, oxytocin stimulates uterine contractions during parturition. It stimulates release of milk from the mammary glands during nursing. It functions in regulating mood and sexual arousal in both males and females. • In birds, it stimulates motility of the oviduct.

5 hormones released by gastrointestinal track with functions 5 marks

Cholecystokinin

Stimulates secretions of pancreas. Stimulates gall bladder contraction

Chymodenin

Stimulates secretion of chymotrypsinogen

Gastrin

Stimulates HCl secretion in stomach

Secretin

Stimulates secretion of bicarbonate

Substance P

Acts as enteric neurotransmitter

Motilin

Increases motility of intestinal villi

Acclimatization 2 marks

An adaptive change within an individual animal; Happens due to chronic exposure to new, naturally occurring environmental conditions.

Example • An animal migrates to a high altitude area • Faces low oxygen partial pressure

Effects • Lung ventilation rate increases initially • Ventilation drops to normal in few days • Animal has acclimatized to the high-altitude

Taste buds and explain taste receptor cells(5max)

The gustatory organs of vertebrates are called taste buds.

A ***taste bud*** is composed of about 50 types of modified epithelial cells, including:

• Supporting cells (sustentacular cells) • Basal cells • Taste receptor cells

Basal cells are progenitor cells that give rise to new taste receptors.

They regularly generate new sensory taste receptor cells which have an active life of only 10 days

Taste Receptor Cells

Outer tips of taste cells are arranged around a minute taste pore.

At the tip of taste cell, several microvilli (taste hairs) protrude into the taste pore.

Microvilli approach the cavity of the mouth. They provide receptor surface for taste. A branching terminal network of taste nerve fibers surrounds the base of taste cells.

Basic components of homeostasis and example of homeostasis, Its mechanism (5max)

Examples of Homeostasis:

Homeostatic regulation of water • Homeostatic regulation of temperature • Regulation of pH • Regulation of glucose concentration • Regulation of osmotic pressure • Regulation of oxygen level • Regulation of ion conc.

Mechanism

Variables

Set point

Living control systems VS physical control systems

Three components: Receptors, Control centre, effectors

Example

• Temperature control system in air conditioners and water heating geysers VS Endothermic animals

Labelled line coading (2max)

Each receptor subtype for taste sensations is connected to a particular set of axons in the nerve.

In such an arrangement, information about one taste e.g. "sweetness" would be carried by some specific subset of axons. Such a pattern is called labeled line coding.

Name glandular cells of thyriod and name their hormones too. (3mqx)

It contains five types of glandular cells that synthesize and secrete six hormones

1-Somatotropes

produce growth hormone (GH)

2-Corticotropes

produce adrenocorticotropic hormone (ACTH)

3-Thyrotropes

Produce thyroid-stimulating hormone (TSH)

4-Gonadotropes

Produce gonadotropic hormones (luteinizing hormone, LH and follicle stimulating hormone, FSH)

5-Lactotropes

Produce prolactin (PRL)

Exocirne gland with 2 examples atleast (5max)

Exocrine glands produce fluid secretions that are delivered through ducts onto the epithelial surfaces of the body. The fluid secretions may be proteins (enzymes) or mucous or both.

Examples:

Salivary glands produce saliva that is delivered to the oral cavity through submandibular and parotid ducts.

Pancreas produces enzyme-containing pancreatic juice that is delivered to the small intestine through pancreatic duct.

Lacrimal glands produce tears that is delivered through lacrimal duct on the surface of eye.

Mammary glands produce milk that is delivered through lactiferous ducts to the nipples

Androgens secreted by adrenal cortex 2 marks

Androgens secreted by adrenal cortex include:
1. Testosterone
2. Dihydrotestosterone
3. Androstenedione
4. Dehydroepiandrosterone (DHEA).

Effects of neuro-modulation 3 marks

- Development of complex behavioral patterns e.g. happiness, exploration, revenge, reward, greed.
- Processes of thinking, cognition, planning, learning and memory
- Behavioral problems: mood swings, sleep disturbances, feelings of stress, anxiety, anger and depression.

2 examples of interoceptive receptors 3 marks

These include:
Receptors of vestibular system that monitor orientation of the body
Thermo-receptors that keep track of thermal state of the body
Chemo-receptors that keep track of chemical state of the body
Proprio-ceptors that monitor position of the muscles and joints
Nociceptors perceive the sensation of pain.
Pacinian corpuscles receive pressure stimulus.

Name and explain hair cell types *5 marks*

Inner hair cells (IHC)

There are about 3,500 IHCs arranged in a straight line or wide U. They are true sensory cells which send impulses via auditory nerve.

Outer hair cells (OHC)

About 12,000-25,000 OHCs in mammals, they are arranged in three or four rows forming a characteristic W shape.

Outer hair cells have both sensory and motor elements. They contribute to hearing sensitivity and frequency selectivity

Hormones of ovary

LH stimulates production of *estrogen* and *progesterone* by ovary.

Explain photoreceptor

Photoreceptors can detect and generate response to a single photon of light. Photoreceptors possess light-sensitive carotenoid pigments retinal and 3dehydroretinal. Carotenoids are associated with opsin proteins to form rhodopsins. Rhodopsins absorb photons of light energy and produce a generator potential

Hair cilia types

Many cilia project from the apical end of each cell (reason of naming). • Cilia are of two types: • Kinocilium • Stereocilia

A hair cell has a single kinocilium. • Kinocilium has a "9 + 2" arrangement of internal microtubules (similar to other motile cilia). • Each hair cell has 20-300nonmotile stereocilia. • Stereocilia are structurally and developmentally distinct from the kinocilium. • They are not formed of microtubules. • They are formed of actin filaments. • The stereocilia are arranged in order of increasing length from one side of the cell to the other.

Diseases by hypothyroidism

It is represented by two types of diseases: • Cretinism • Goiter

Cretinism results from iodine deficiency during early stages of development

Inadequate production of thyroxine in adults leads to ***goiter***.

active potential phases

Stigma and ocelli 3

Eyespot or Stigma

Simplest photoreceptive structure. Found in some protozoa e.g. euglena. It is a bright red colored organelle and has carotenoid pigments. It gives a sense of light and dark. Helps in phototaxis.

Eyecups or Ocelli

Multi-cellular photoreceptive structure that consists of a cuplike depression containing photoreceptor cells; Found in cnidarians and flatworms e.g. Planaria.

Cannot form image, provides the animal a sense of direction only

Donnan equilibrium 2

The phenomenon of unequal distribution of ions across the two sides of a differentially permeable membrane is called Donnan Equilibrium.

Discovered by Frederick Donnan in 1911.

(10 Marks 2 Questions)

Prostaglandins

Prostaglandins are about 16 hormone-like local regulators substances .

They are derived from cyclic, long-chain, unsaturated, hydroxy fatty acids.

Synthesis

Prostaglandins are produced by all or nearly all tissues.

They are synthesized in membranes from arachidonic acid, which is produced by cleavage of membrane phospholipids by phospholipases.

Mode of Action

In some cases, they act locally as paracrine agents.

In other cases they act on distant target tissues in endocrine fashion.

They have a rapid, shortlasting effect, similar to that of lipid-insoluble hormones.

They bind to cell-surface receptors linked to the cAMP pathway.

Functions

Prostaglandins have diverse actions on a variety of tissues, particularly involving smooth muscles.

Prostaglandins Role in Fertilization

Prostaglandins present in semen aid in fertilization by reacting with the female cervical mucus to make it more receptive to sperm movement.

They also stimulate the smooth muscles of the uterine wall to contract, helping sperms reach an egg.

Prostaglandins Aid in Labor

Certain prostaglandins are secreted by the placenta at the onset of childbirth.

They make the muscles of uterus more excitable, enhancing uterine contractions during labor.

Prostaglandins Role in Immune Response

Damaged tissues produce prostaglandins during immune response.

Some prostaglandins act as local regulators of inflammation.

They promote fever and inflammation and also intensify the sensation of pain.

Prostaglandins Role in Immune Response

They worsen pain by increasing nociceptor sensitivity.

Prostaglandin E2, formed from interleukin-1, produced in immune response, causes fever by acting on the hypothalamus to elicit fever reaction.

The anti-pyretic, anti-inflammatory and pain relieving effects of aspirin and ibuprofen are actually due to the inhibition of prostaglandin synthesis by these drugs.

Prostaglandins Role in Blood Clot Formation

Prostaglandins help platelets to aggregate and form blood clots.

This is the basis of use of aspirin by patients at risk for a heart attack due to formation of clots.

Maintenance of Stomach Lining

Prostaglandins help to maintain a protective lining in the stomach.

As aspirin interferes with prostaglandin synthesis, long-term aspirin therapy can result in damage to this protective lining, causing stomach irritation and ulcer.

Production of Erythropoietin

Several prostaglandins stimulate the production of erythropoietin by kidneys.

Erythropoietin is a hormone that stimulates the production of erythrocytes.

Blood Pressure Regulation

Some prostaglandins act on the smooth muscles of blood vessels.

They regulate blood pressure by vasodilation and vasoconstriction.

Functions of T3 and T4 and Hyperthyroidism

These hormones have diverse effects on the physiology of virtually all tissues of the body.

Their roles can be grouped into two categories:

Metabolic roles

Developmental roles

Metabolic Roles of T3 & T4

They stimulate cellular respiration, oxygen consumption and metabolic rate and, in turn, heat production. (important in thermoregulation).

They sensitize some tissues to epinephrine that helps to maintain normal blood pressure, heart rate and muscle tone.

They also promote normal motility of gastrointestinal tract.

They also regulate reproductive functions.

Developmental Roles of T3 & T4

They significantly affect the development and maturation of vertebrate animals.

They are involved in the normal functioning of bone-forming cells and the branching of nerve cells during embryonic development of the brain.

They control metamorphosis of a tadpole larva into adult frog.

Developmental effects of growth hormone occur only in the presence of thyroid hormones.

Hyperthyroidism

Excessive secretion of thyroid hormones as in Graves' disease,

It is an autoimmune disorder in which antibodies that mimic TSH bind to the receptors for TSH and cause sustained thyroxine production.

It leads to:

High body temp

Profuse sweating

Weight loss

Irritability

high blood pressure

protruding eyes (exophthalmia)